

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Monday 23 June 2025

Afternoon (Time: 1 hour 30 minutes) **Paper reference** **9FM0/4A**

Further Mathematics

Advanced

PAPER 4A: Further Pure Mathematics 2

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P75689A

©2025 Pearson Education Ltd.
Y:1/1/1/




Pearson

1. The set $S = \{1, 3, 5, 9, 11, 13\}$ forms the group G , under the operation multiplication modulo 14

(a) Complete the Cayley table below for the group G

$\times_{14}$	1	3	5	9	11	13
1	1	3	5	9	11	13
3	3	9	1	13	5	11
5	5	1	11			
9	9	13		11		
11	11	5			9	
13	13	11				1

A spare table can be found on page 5 if you need to rewrite your Cayley table.

(3)

(b) Write down a subgroup of G of order 2

(1)

The group H is defined by the Cayley table below.

$*$	p	q	r	s	t	u
p	p	q	r	s	t	u
q	q	t	u	r	s	p
r	r	u	t	q	p	s
s	s	r	q	p	u	t
t	t	s	p	u	r	q
u	u	p	s	t	q	r

(c) Show that G and H are isomorphic.

(3)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 1 continued

Only use this grid if you need to rewrite your Cayley table.

$\times_{14}$	1	3	5	9	11	13
1	1	3	5	9	11	13
3	3	9	1	13	5	11
5	5	1	11			
9	9	13		11		
11	11	5			9	
13	13	11				1

(Total for Question 1 is 7 marks)



2.

$$\mathbf{A} = \begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix}$$

(a) Determine the eigenvalues of matrix $\mathbf{A}$

(3)

(b) Hence determine an orthogonal matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{D} = \mathbf{P}^T \mathbf{A} \mathbf{P}$$

(5)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 8 marks)



3. The circle C has equation

$$|z - 5 + 4i| = 2|z - 2 + i|$$

where $z = x + iy$

(a) Show that an equation for C is

$$x^2 - 2x + y^2 - 7 = 0 \tag{3}$$

The half-line with equation

$$\arg(z-a) = -\frac{3\pi}{4}$$

where a is a real constant, is a tangent to C

(b) Determine the value of a . (6)

DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 9 marks)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 9 marks)



5. (i) A password is to be created from 5 characters using the following criteria

- one capital letter
- 3 distinct single digits
- one of the four symbols @ £ \$ &

Determine how many different passwords can be created if

- (a) the characters are in the order of the bullet points above, (2)

- (b) the characters can be in any order. (1)

- (ii) Use Fermat's Little Theorem to determine the least positive residue of 4^{50} modulo 13

- (iii) Determine the total number of different, positive **odd** numbers, less than 1000, that contain the digit 2
- (4)**

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

DO NOT WRITE IN THIS AREA

Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 10 marks)



6.

$$I_n = \int_0^2 (4 - x^2)^n dx$$

(a) Prove that, for $n \geq 1$

$$I_n = \frac{8n}{2n+1} I_{n-1} \quad (4)$$

(b) Using the result from part (a) and showing all stages of your working, determine the value of n for which

$$I_n = \frac{65536}{315}$$

(Solutions relying entirely on calculator technology are not acceptable.)

(4)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 8 marks)



7. (i) Explain why the congruence equation

$$21x \equiv 11 \pmod{36}$$

has no solutions.

(2)

- (ii) Use modular arithmetic to solve the congruence equation

$$36y \equiv 192 \pmod{40}$$

(4)



DO NOT WRITE IN THIS AREA

Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 6 marks)



8.

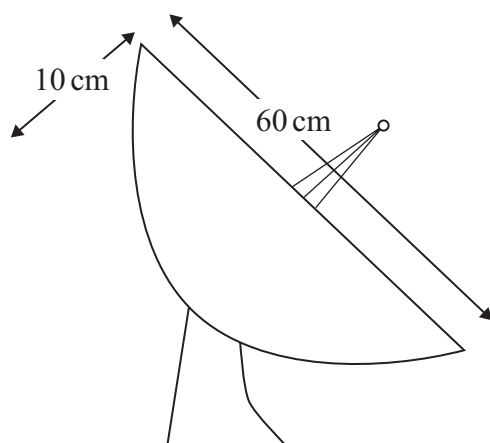


Figure 1

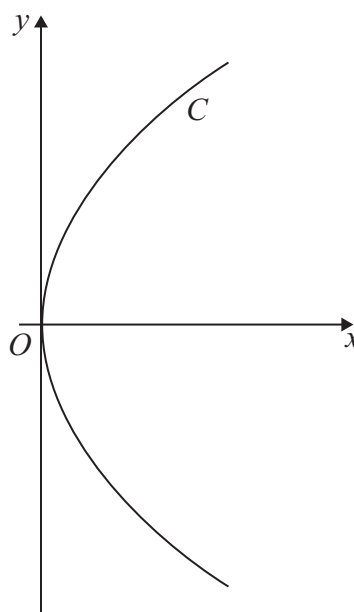


Figure 2

Figure 1 shows a satellite dish.

Figure 2 shows a sketch of the curve C with equation

$$y^2 = Ax \quad 0 \leq x \leq 10$$

where A is a positive constant.

The curved inner surface of the satellite dish is modelled by the surface of revolution formed by rotating curve C through π radians about the x -axis.

The inner surface of the satellite dish has

- a largest diameter of 60 cm
- a depth of 10 cm

as shown in Figure 1.

- Determine the value of A . (2)
- Using algebraic integration, determine, in cm^2 , the area of the curved inner surface of the satellite dish, according to the model. Give your answer to 2 significant figures. (7)
- State a limitation of the model. (1)



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 10 marks)



9. The transformation T from the z -plane to the w -plane, where $z = x + iy$ and $w = u + iv$ is given by

$$w = \frac{z}{z + 2i} \quad z \neq -2i$$

The circle with equation $|z| = 4$ is mapped by T onto the circle C .

Determine

- (i) the coordinates of the centre of C
- (ii) the radius of C

(8)



DO NOT WRITE IN THIS AREA

Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 9 is 8 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information	
Candidate surname	Other names
Centre Number	Candidate Number
Pearson Edexcel Level 3 GCE	
Monday 24 June 2024	
Afternoon (Time: 1 hour 30 minutes)	Paper reference 9FM0/4A
Further Mathematics Advanced PAPER 4A: Further Pure Mathematics 2	
You must have: Mathematical Formulae and Statistical Tables (Green), calculator	Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P72800A

©2024 Pearson Education Ltd.
F:1/1/1/1/1/



1. In this question you must show detailed reasoning.

Use Fermat's Little Theorem to determine the least positive residue of

$$21^{80} \pmod{23}$$

(4)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2. Determine a closed form for the recurrence system

$$\begin{aligned} u_1 &= 4 & u_2 &= 6 \\ u_{n+2} &= 6u_{n+1} - 9u_n & (n &= 1, 2, 3, \dots) \end{aligned} \quad (5)$$

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)



3.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

- (a) Use the Euclidean Algorithm to determine the highest common factor h of 234 and 96

(3)

- (b) Hence determine integers a and b such that

$$234a + 96b = h$$

(3)

- (c) Solve the congruence equation

$$96x \equiv 36 \pmod{234}$$

(5)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 11 marks)



4.

$$\mathbf{A} = \begin{pmatrix} 4 & 2 & 0 \\ 2 & p & -2 \\ 0 & -2 & 2 \end{pmatrix} \quad \text{where } p \text{ is a constant}$$

Given that $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ is an eigenvector of $\mathbf{A}$,

- (a) determine the eigenvalue corresponding to this eigenvector. (2)
- (b) Hence show that $p = 3$ (1)
- (c) Determine
 - (i) the remaining eigenvalues of $\mathbf{A}$,
 - (ii) corresponding eigenvectors for these eigenvalues. (6)
- (d) Hence determine a matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that $\mathbf{A} = \mathbf{PDP}^T$ (3)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 12 marks)



5. (i) A circle C in the complex plane is defined by the locus of points satisfying

$$|z - 3i| = 2|z|$$

- (a) Determine a Cartesian equation for C , giving your answer in simplest form. (3)

- (b) On an Argand diagram, shade the region defined by

$$\{z \in \mathbb{C} : |z - 3i| > 2|z|\}$$

(2)

- (ii) The transformation T from the z -plane to the w -plane is given by

$$w = z^3$$

- (a) Describe the geometric effect of T . (2)

The region R in the z -plane is given by

$$\left\{ z \in \mathbb{C} : 0 < \arg z < \frac{\pi}{4} \right\}$$

- (b) On a **different** Argand diagram, sketch the image of R under T . (2)



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 9 marks)

6.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

$$I_n = \int \frac{\cos(nx)}{\sin x} dx \quad n \geq 1$$

(a) Show that, for $n \geq 1$

$$I_{n+2} = \frac{2 \cos(n+1)x}{n+1} + I_n \quad (6)$$

(b) Hence determine the exact value of

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos(5x)}{\sin x} dx$$

giving the answer in the form $a + b \ln c$ where a , b and c are rational numbers to be found.

(5)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 11 marks)



7. The set of matrices $G = \{\mathbf{I}, \mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{E}\}$ where

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \quad \mathbf{C} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \mathbf{D} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \quad \mathbf{E} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

with the operation $\otimes_2$ of matrix multiplication with entries evaluated modulo 2, forms a group.

- (a) Show that $\mathbf{B}$ is an element of order 3 in G . (2)
- (b) Determine the orders of the other elements of G . (3)
- (c) Give a reason why G is **not** isomorphic to
 - (i) a cyclic group of order 6
 - (ii) the group of symmetries of a regular hexagon. (2)

The group H of permutations of the numbers 1, 2 and 3 contains the following elements, denoted in two-line notation,

$$\begin{array}{lll} e = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} & a = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} & b = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \\ c = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} & d = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} & f = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \end{array}$$

- (d) Determine an isomorphism between the groups G and H . (3)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 10 marks)



8.



Figure 1

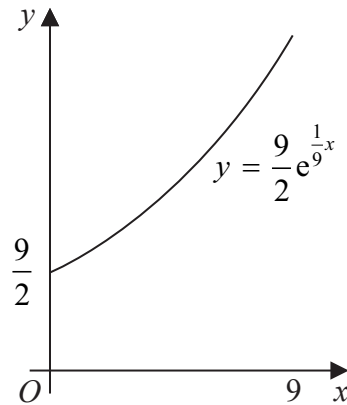


Figure 2

Figure 1 shows a French horn with a detachable bell section.

The shape of the bell section can be modelled by rotating an exponential curve through 360° about the x -axis, where units are centimetres.

The model uses the curve shown in Figure 2, with equation

$$y = \frac{9}{2}e^{\frac{1}{9}x} \quad 0 \leq x \leq 9$$

- (a) Show that, according to this model, the external surface area of the bell section is given by

$$K \int_0^9 e^{\frac{1}{9}x} \sqrt{4 + e^{\frac{2}{9}x}} \, dx$$

where K is a real constant to be determined.

(3)

- (b) Use the substitution $u = e^{\frac{1}{9}x}$ to show that

$$\int_0^9 e^{\frac{1}{9}x} \sqrt{4 + e^{\frac{2}{9}x}} \, dx = 9 \int_a^b \frac{2u + u^3}{\sqrt{4u^2 + u^4}} \, du + 18 \int_a^b \frac{1}{\sqrt{4 + u^2}} \, du$$

where a and b are constants to be determined.

(5)

Hence, using algebraic integration,

- (c) determine, according to the model, the external surface area of the bell section of the horn, giving your answer to 3 significant figures.

(5)



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 is 13 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Monday 26 June 2023

Afternoon (Time: 1 hour 30 minutes) **Paper reference** **9FM0/4A**

Further Mathematics

Advanced

PAPER 4A: Further Pure Mathematics 2

You must have:
Mathematical Formulae and Statistics Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 10 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P74083A

©2022 Pearson Education Ltd.
N:1/1/1/1/1/




Pearson

1.

$$\mathbf{A} = \begin{pmatrix} -1 & a \\ 3 & 8 \end{pmatrix}$$

where a is a constant.

(a) Determine, in expanded form in terms of a , the characteristic equation for $\mathbf{A}$. (2)

(b) Hence use the Cayley-Hamilton theorem to determine values of a and b such that

$$\mathbf{A}^3 = \mathbf{A} + b\mathbf{I}$$

where $\mathbf{I}$ is the 2×2 identity matrix. (4)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. A complex number z is represented by the point P in the complex plane.

Given that z satisfies

$$|z - 6| = 2|z + 3i|$$

- (a) show that the locus of P passes through the origin and the points -4 and $-8i$ (2)

- (b) Sketch on an Argand diagram the locus of P as z varies. (2)

- (c) On your sketch, shade the region which satisfies both

$$|z - 6| \geq 2|z + 3i| \quad \text{and} \quad |z| \leq 4 \quad (2)$$



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 6 marks)



3. In a model for the number of subscribers to a new social media channel it is assumed that
- each week 20% of the subscribers at the start of the week cancel their subscriptions
 - between the start and end of week n the channel gains $20n$ new subscribers

Given that at the end of week 1 there were 25 subscribers,

- (a) explain why the number of subscribers at the end of week n , U_n , is modelled by the recurrence relation

$$U_1 = 25 \quad U_{n+1} = 0.8U_n + 20(n+1) \quad n = 1, 2, 3, \dots \quad (2)$$

- (b) Prove by induction that for $n \geq 1$

$$U_n = 325 \left(\frac{4}{5} \right)^{n-1} + 100n - 400 \quad (5)$$

Given that 6 months after starting the channel there were approximately 1800 subscribers,

- (c) evaluate the model in the light of this information. (2)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 9 marks)



- (2)

- (3)

- (1)

- (3)

DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4 continued.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 9 marks)



5. (i) A security code is made up of 4 numerical digits followed by 3 **distinct** uppercase letters.

Given that the digits must be from the set $\{1, 2, 3, 4, 5\}$ and the letters from the set $\{A, B, C, D\}$

- (a) determine the total number of possible codes using this system.

To enable more codes to be generated, the system is adapted so that the 3 letters can appear anywhere in the code **but** no letter can be next to another letter.

- (b) Determine the increase in the number of codes using this adapted system.

(4)

- (ii) A combination lock code consists of four **distinct** digits that can be read as a positive integer, $N = abcd$, satisfying

- all the digits are odd
- N is divisible by 9
- the digits appear in either ascending or descending order
- $N \equiv e \pmod{ab}$ where ab is read as a two-digit number and e is the odd digit that is not used in the code

- (a) Use the first two properties to determine the four digits used in the code.

- (b) Hence determine the code on the lock.

(4)



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 8 marks)



6. Determine a closed form for the recurrence relation

$$\begin{aligned} u_0 &= 1 & u_1 &= 4 \\ u_{n+2} &= 2u_{n+1} - \frac{4}{3}u_n + n & n &\geq 0 \end{aligned} \quad (7)$$



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 7 marks)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 6 marks)



8.

$$I_n = \int_0^2 (x-2)^n e^{4x} dx \quad n \geq 0$$

(a) Prove that for $n \geq 1$

$$I_n = -a^{n-2} - \frac{n}{4} I_{n-1}$$

where a is a constant to be determined.

(4)

(b) Hence determine the exact value of

$$\int_0^2 (x-2)^2 e^{4x} dx$$

(3)



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 7 marks)



9.

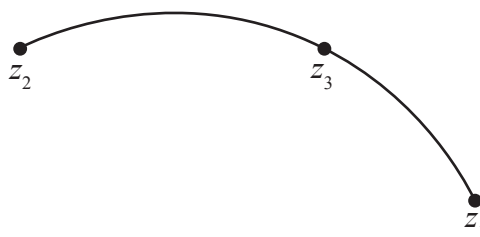
**Figure 1**

Figure 1 shows a locus in the complex plane.

The locus is an arc of a circle from the point represented by $z_1 = 3 + 2i$ to the point represented by $z_2 = a + 4i$, where a is a constant, $a \neq 1$

Given that

- the point $z_3 = 1 + 4i$ also lies on the locus
- the centre of the circle has real part equal to -1

(a) determine the value of a .

(2)

(b) Hence determine a complex equation for the locus, giving any angles in the equation as positive values.

(3)



DO NOT WRITE IN THIS AREA

Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 5 marks)



10.

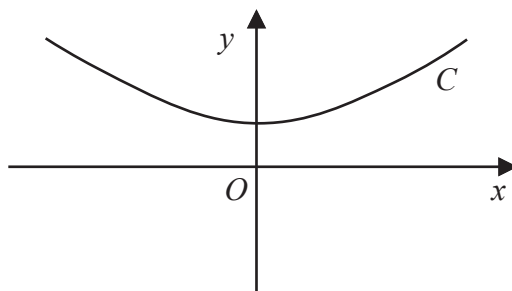


Figure 2

A solid playing piece for a board game is modelled by rotating the curve C , shown in Figure 2, through 2π radians about the x -axis.

The curve C has equation

$$y = \sqrt{1 + \frac{x^2}{9}} \quad -4 \leq x \leq 4$$

with units as centimetres.

- (a) Show that the total surface area, $S \text{ cm}^2$, of the playing piece is given by

$$S = p\pi \int_{-4}^4 \sqrt{81 + 10x^2} \, dx + q\pi$$

where p and q are constants to be determined.

(6)

Using the substitution $x = \frac{9}{\sqrt{10}} \sinh u$, or another algebraic integration method, and showing all your working,

- (b) determine the total surface area of the playing piece, giving your answer to the nearest cm^2

(6)

[illegible]

DO NOT WRITE IN THIS AREA

Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 10 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Candidate Number

--	--	--	--	--

--	--	--	--

Pearson Edexcel Level 3 GCE

Time 1 hour 30 minutes

Paper
reference

9FM0/4A

Further Mathematics

Advanced

PAPER 4A: Further Pure Mathematics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 10 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P65506A

©2022 Pearson Education Ltd.

1/1/1/1/



Pearson

1. The group S_4 is the set of all possible permutations that can be performed on the four numbers 1, 2, 3 and 4, under the operation of composition.

For the group S_4

- (a) write down the identity element, (1)

- (b) write down the inverse of the element a , where

$$a = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad (1)$$

- (c) demonstrate that the operation of composition is associative using the following elements

$$a = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \quad \text{and} \quad c = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \quad (2)$$

- (d) Explain why it is possible for the group S_4 to have a subgroup of order 4
You do not need to find such a subgroup. (2)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. Matrix $\mathbf{M}$ is given by

$$\mathbf{M} = \begin{pmatrix} 1 & 0 & a \\ -3 & b & 1 \\ 0 & 1 & a \end{pmatrix}$$

where a and b are integers, such that $a < b$

Given that the characteristic equation for $\mathbf{M}$ is

$$\lambda^3 - 7\lambda^2 + 13\lambda + c = 0$$

where c is a constant,

(a) determine the values of a , b and c .

(5)

(b) Hence, using the Cayley–Hamilton theorem, determine the matrix $\mathbf{M}^{-1}$

(3)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 8 marks)



3.

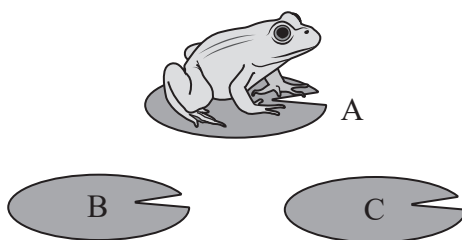


Figure 1

There are three lily pads on a pond. A frog hops repeatedly from one lily pad to another.

The frog starts on lily pad A, as shown in Figure 1.

In a model, the frog hops from its position on one lily pad to either of the other two lily pads with equal probability.

Let p_n be the probability that the frog is on lily pad A after n hops.

- (a) Explain, with reference to the model, why $p_1 = 0$ (1)

The probability p_n satisfies the recurrence relation

$$p_{n+1} = \frac{1}{2}(1 - p_n) \quad n \geq 1 \quad \text{where } p_1 = 0$$

- (b) Prove by induction that, for $n \geq 1$

$$p_n = \frac{2}{3} \left(-\frac{1}{2} \right)^n + \frac{1}{3} \quad (6)$$

- (c) Use the result in part (b) to explain why, in the long term, the probability that the frog is on lily pad A is $\frac{1}{3}$ (1)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 8 marks)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

4. (a) Use the Euclidean algorithm to show that 124 and 17 are relatively prime (coprime). (2)

(b) Hence solve the equation

$$124x + 17y = 10 \tag{3}$$

(c) Solve the congruence equation

$$124x \equiv 6 \pmod{17} \tag{2}$$



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 7 marks)



5. The locus of points z satisfies

$$|z + ai| = 3|z - a|$$

where a is an integer.

The locus is a circle with its centre in the third quadrant and radius $\frac{3}{2}\sqrt{2}$

Determine

(a) the value of a , (4)

(b) the coordinates of the centre of the circle. (2)

Question 5 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 5 is 6 marks)



P 6 5 5 0 6 A 0 1 5 3 2

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

6. (a) Determine the general solution of the recurrence relation

$$u_n = 2u_{n-1} - u_{n-2} + 2^n \quad n \geq 2 \quad (4)$$

- (b) Hence solve this recurrence relation given that $u_0 = 2u_1$ and $u_4 = 3u_2$ (2)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 6 marks)



7. (i) The polynomial $F(x)$ is a quartic such that

$$F(x) = px^4 + qx^3 + 2x^2 + rx + s$$

where p, q, r and s are distinct constants.

Determine the number of possible quartics given that

- (a) the constants p, q, r and s belong to the set $\{-4, -2, 1, 3, 5\}$ (1)
- (b) the constants p, q, r and s belong to the set $\{-4, -2, 0, 1, 3, 5\}$ (1)

- (ii) A 3-digit positive integer $N = abc$ has the following properties

- N is divisible by 11
- the sum of the digits of N is even
- $N \equiv 8 \pmod{9}$

- (a) Use the first two properties to show that

$$a - b + c = 0 \tag{2}$$

- (b) Hence determine all possible integers N , showing all your working and reasoning. (4)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 8 marks)



8. The locus of points $z = x + iy$ that satisfy

$$\arg\left(\frac{z - 8 - 5i}{z - 2 - 5i}\right) = \frac{\pi}{3}$$

is an arc of a circle C .

- (a) On an Argand diagram sketch the locus of z . (2)
- (b) Explain why the centre of C has x coordinate 5 (1)
- (c) Determine the radius of C . (2)
- (d) Determine the y coordinate of the centre of C . (2)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 7 marks)



9.

$$I_n = \int_0^{\frac{\pi}{2}} \sin^n 2x \, dx$$

(a) Prove that for $n \geq 2$

$$I_n = \frac{n-1}{n} I_{n-2} \quad (4)$$

(b) Hence determine the exact value of

$$\int_0^{\frac{\pi}{2}} 64 \sin^5 x \cos^5 x \, dx \quad (3)$$



DO NOT WRITE IN THIS AREA

Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 7 marks)



10.

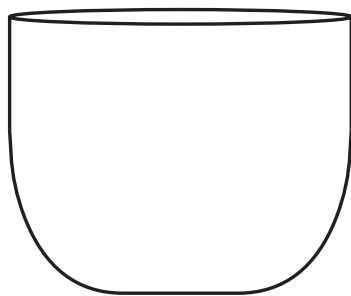


Figure 2

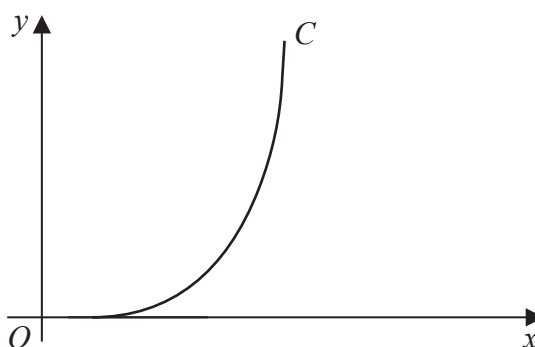


Figure 3

Figure 2 shows a picture of a plant pot.

The plant pot has

- a flat circular base of radius 10 cm
- a height of 15 cm

Figure 3 shows a sketch of the curve C with parametric equations

$$x = 10 + 15t - 5t^3 \quad y = 15t^2 \quad 0 \leq t \leq 1$$

The curved inner surface of the plant pot is modelled by the surface of revolution formed by rotating curve C through 2π radians about the y -axis.

- (a) Show that, according to the model, the area of the curved inner surface of the plant pot is given by

$$150\pi \int_0^1 (2 + 3t + 2t^2 + 2t^3 - t^5) dt \quad (5)$$

- (b) Determine, according to the model, the total area of the inner surface of the plant pot.

(4)

Each plant pot will be painted with one coat of paint, both inside and outside. The paint in one tin will cover an area of 12 m^2

- (c) Use the answer to part (b) to estimate how many plant pots can be painted using one tin of paint.

(2)

- (d) Give a reason why the model might not give an accurate answer to part (c).

(1)



DO NOT WRITE IN THIS AREA

Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 10 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Time 1 hour 30 minutes	Paper reference	9FM0/4A	
Further Mathematics Advanced PAPER 4A: Further Pure Mathematics 2			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ►

P66802A

©2021 Pearson Education Ltd.

1/1/1/



1. In this question you must show detailed reasoning.

Without performing any division, explain why $n = 20210520$ is divisible by 66

(4)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 4 marks)



2. A binary operation $\star$ on the set of non-negative integers, $\mathbb{Z}_0^+$, is defined by

$$m \star n = |m - n| \quad m, n \in \mathbb{Z}_0^+$$

- (a) Explain why $\mathbb{Z}_0^+$ is closed under the operation $\star$ (1)
- (b) Show that 0 is an identity for $(\mathbb{Z}_0^+, \star)$ (2)
- (c) Show that all elements of $\mathbb{Z}_0^+$ have an inverse under $\star$ (2)
- (d) Determine if $\mathbb{Z}_0^+$ forms a group under $\star$, giving clear justification for your answer. (3)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 8 marks)



3. (a) Use the Euclidean Algorithm to find integers a and b such that

$$125a + 87b = 1 \quad (5)$$

- (b) Hence write down a multiplicative inverse of 87 modulo 125 (1)

- (c) Solve the linear congruence

$$87x \equiv 16 \pmod{125} \quad (2)$$



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

4. Let G be a group of order $46^{46} + 47^{47}$

Using Fermat's Little Theorem and explaining your reasoning, determine which of the following are possible orders for a subgroup of G

(i) 11

(ii) 21

(7)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



5. The point P in the complex plane represents a complex number z such that

$$|z + 9| = 4|z - 12i|$$

Given that, as z varies, the locus of P is a circle,

- (a) determine the centre and radius of this circle.

(6)

- (b) Shade on an Argand diagram the region defined by the set

$$\{z \in \mathbb{C} : |z + 9| < 4|z - 12i|\} \cap \left\{z \in \mathbb{C} : -\frac{\pi}{4} < \arg\left(z - \frac{3 + 44i}{5}\right) < \frac{\pi}{4}\right\}$$

(4)



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 10 marks)



6. A recurrence system is defined by

$$u_{n+2} = 9(n+1)^2 u_n - 3u_{n+1} \quad n \geq 1$$

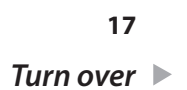
$$u_1 = -3, u_2 = 18$$

Prove by induction that, for $n \in \mathbb{N}$,

$$u_n = (-3)^n n! \quad (6)$$



DO NOT WRITE IN THIS AREA



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 6 marks)



7.

In this question you must show all stages of your working.

You must not use the integration facility on your calculator.

$$I_n = \int t^n \sqrt{4 + 5t^2} \, dt \quad n \geq 0$$

(a) Show that, for $n > 1$

$$I_n = \frac{t^{n-1}}{5(n+2)} (4 + 5t^2)^{\frac{3}{2}} - \frac{4(n-1)}{5(n+2)} I_{n-2} \quad (5)$$

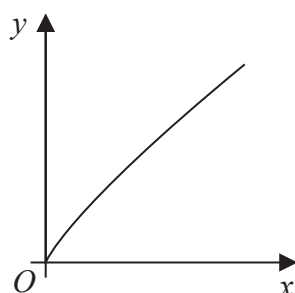


Figure 1

The curve shown in Figure 1 is defined by the parametric equations

$$x = \frac{1}{\sqrt{5}} t^5 \quad y = \frac{1}{2} t^4 \quad 0 \leq t \leq 1$$

This curve is rotated through 2π radians about the x -axis to form a hollow open shell.

(b) Show that the external surface area of the shell is given by (5)

$$\pi \int_0^1 t^7 \sqrt{4 + 5t^2} \, dt$$

Using the results in parts (a) and (b) and making each step of your working clear,

(c) determine the value of the external surface area of the shell, giving your answer to 3 significant figures.

(5)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 7 continued

Handwriting practice area with horizontal lines.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing answers.

(Total for Question 7 is 15 marks)



8.

$$\mathbf{A} = \begin{pmatrix} 5 & -2 & 5 \\ 0 & 3 & p \\ -6 & 6 & -4 \end{pmatrix} \quad \text{where } p \text{ is a constant}$$

Given that $\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ is an eigenvector for $\mathbf{A}$

(a) (i) determine the eigenvalue corresponding to this eigenvector (1)

(ii) hence show that $p = 2$ (2)

(iii) determine the remaining eigenvalues and corresponding eigenvectors of $\mathbf{A}$ (7)

(b) Write down a matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that $\mathbf{A} = \mathbf{PDP}^{-1}$ (1)

(c) (i) Solve the differential equation $\dot{u} = ku$, where k is a constant. (2)

With respect to a fixed origin O , the velocity of a particle moving through space is modelled by

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{pmatrix} = \mathbf{A} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

By considering $\begin{pmatrix} u \\ v \\ w \end{pmatrix} = \mathbf{P}^{-1} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ so that $\begin{pmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{pmatrix} = \mathbf{P}^{-1} \begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{pmatrix}$

(ii) determine a general solution for the displacement of the particle. (4)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 8 continued

Handwriting practice area with horizontal lines.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 8 continued

Handwriting practice area with horizontal lines.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 is 17 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information			
Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Friday 23 October 2020			
Afternoon (Time: 1 hour 30 minutes)		Paper Reference 9FM0/4A	
Further Mathematics Advanced Paper 4A: Further Pure Mathematics 2			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B)
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P62681A

©2020 Pearson Education Ltd.

1/1/1/1/1/



1. A small sports club has 12 adult members and 14 junior members.

The club needs to enter a team of 8 players for a particular competition.

Determine the number of ways in which the team can be selected if

- (i) there are no restrictions on the team,

(1)

- (ii) the team must contain 4 adults and 4 juniors,

(2)

- (iii) more than half the team must be adults.

(3)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. Solve the recurrence system

$$\begin{aligned} u_1 &= 1 & u_2 &= 4 \\ 9u_{n+2} - 12u_{n+1} + 4u_n &= 3n \end{aligned}$$

(9)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 9 marks)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

3.

$$\mathbf{M} = \begin{pmatrix} 1 & k & -2 \\ 2 & -4 & 1 \\ 1 & 2 & 3 \end{pmatrix}$$

where k is a constant.

(a) Show that, in terms of k , a characteristic equation for $\mathbf{M}$ is given by

$$\lambda^3 - (2k + 13)\lambda + 5(k + 6) = 0 \quad (3)$$

Given that $\det \mathbf{M} = 5$

(b) (i) find the value of k

(ii) use the Cayley-Hamilton theorem to find the inverse of $\mathbf{M}$. (7)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 10 marks)



4.

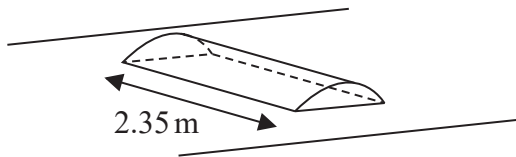


Figure 1

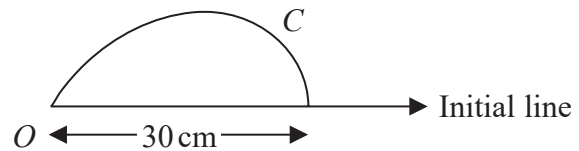


Figure 2

Figure 1 shows a sketch of a design for a road speed bump of width 2.35 metres. The speed bump has a uniform cross-section with vertical ends and its length is 30 cm. A side profile of the speed bump is shown in Figure 2.

The curve C shown in Figure 2 is modelled by the polar equation

$$r = 30(1 - \theta^2) \quad 0 \leq \theta \leq 1$$

The units for r are centimetres and the initial line lies along the road surface, which is assumed to be horizontal.

Once the speed bump has been fixed to the road, the visible surfaces of the speed bump are to be painted.

Determine, in cm^2 , the area that is to be painted, according to the model.

(10)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 10 marks)



5. A transformation T from the z -plane to the w -plane is given by

$$w = \frac{1 - 3z}{z + 2i} \quad z \neq -2i$$

The circle with equation $|z + i| = 3$ is mapped by T onto the circle C .

- (a) Show that the equation for C can be written as

$$3|w + 3| = |1 + (3 - w)i| \quad (4)$$

- (b) Hence find

- (i) a Cartesian equation for C ,
(ii) the centre and radius of C .

(6)



DO NOT WRITE IN THIS AREA



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA



6.

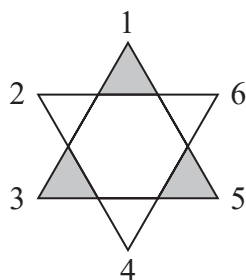


Figure 3

Figure 3 shows a plane shape made up of a regular hexagon with an equilateral triangle joined to each edge and with alternate equilateral triangles shaded.

The symmetries of this shape are the rotations and reflections of the plane that preserve the shape and its shading.

The symmetries of the shape can be represented by permutations of the six vertices labelled 1 to 6 in Figure 3. The set of these permutations with the operation of composition form a group, G .

- (a) Describe geometrically the symmetry of the shape represented by the permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 5 & 6 & 1 & 2 \end{pmatrix}$$

(2)

- (b) Write down, in similar two-line notation, the remaining elements of the group G .

(4)

- (c) Explain why each of the following statements is false, making your reasoning clear.

(i) G has a subgroup of order 4

(ii) G is cyclic.

(2)

Diagram 1, on page 23, shows an unshaded shape with the same outline as the shape in Figure 3.

- (d) Shade the shape in Diagram 1 in such a way that the group of symmetries of the resulting shaded shape is isomorphic to the cyclic group of order 6

(2)



DO NOT WRITE IN THIS AREA



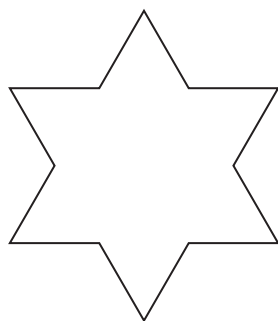
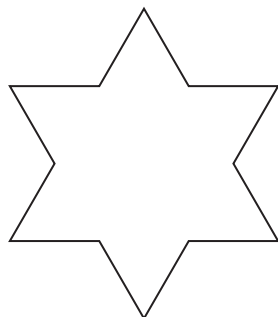
Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 6 continued**Diagram 1****Spare copy of Diagram 1**

Only use this diagram if you need to redraw your answer to part (d).

(Total for Question 6 is 10 marks)

7.

$$I_n = \int (4 - x^2)^{-n} dx \quad n > 0$$

(a) Show that, for $n > 0$

$$I_{n+1} = \frac{x}{8n(4 - x^2)^n} + \frac{2n - 1}{8n} I_n \quad (5)$$

(b) Find I_2 (3)

DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA



8. The four digit number $n = abcd$ satisfies the following properties:

- (1) $n \equiv 3 \pmod{7}$
- (2) n is divisible by 9
- (3) the first two digits have the same sum as the last two digits
- (4) the digit b is smaller than any other digit
- (5) the digit c is even

(a) Use property (1) to explain why $6a + 2b + 3c + d \equiv 3 \pmod{7}$ (2)

(b) Use properties (2), (3) and (4) to show that $a + b = 9$ (4)

(c) Deduce that $c \equiv 5(a - 1) \pmod{7}$ (2)

(d) Hence determine the number n , verifying that it is unique. You must make your reasoning clear. (4)



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information			
Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Tuesday 25 June 2019			
Morning (Time: 1 hour 30 minutes)		Paper Reference 9FM0/4A	
Further Mathematics Advanced Paper 4A: Further Pure Mathematics 2			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B)
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P61183A

©2019 Pearson Education Ltd.

1/1/1



Answer ALL questions. Write your answers in the spaces provided.

1. A complex number $z = x + iy$ is represented by the point P in an Argand diagram.

Given that

$$|z - 3| = 4|z + 1|$$

- (a) show that the locus of P has equation

$$15x^2 + 15y^2 + 38x + 7 = 0$$

(2)

- (b) Hence find the maximum value of $|z|$

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 5 marks)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 11 marks)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 8 marks)



4. (i) Use Fermat's Little Theorem to find the least positive residue of 6^{542} modulo 13 (5)
- (ii) Seven students, Alan, Brenda, Charles, Devindra, Enid, Felix and Graham, are attending a concert and will sit in a particular row of 7 seats. Find the number of ways they can be seated if
- (a) there are no restrictions where they sit in the row, (1)
- (b) Alan, Enid, Felix and Graham sit together, (2)
- (c) Brenda sits at one end of the row and Graham sits at the other end of the row, (2)
- (d) Charles and Devindra do not sit together. (2)



DO NOT WRITE IN THIS AREA

[illegible]

5.

$$I_n = \int \operatorname{cosec}^n x \, dx \quad n \in \mathbb{Z}$$

(a) Prove that, for $n \geq 2$

$$I_n = \frac{n-2}{n-1} I_{n-2} - \frac{\operatorname{cosec}^{n-2} x \cot x}{n-1} \quad (4)$$

(b) Hence show that

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \operatorname{cosec}^6 x \, dx = \frac{56}{135} \sqrt{3} \quad (4)$$



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 8 marks)



6. (i) A binary operation $*$ is defined on positive real numbers by

$$a * b = a + b + ab$$

Prove that the operation $*$ is associative.

(4)

- (ii) The set $G = \{1, 2, 3, 4, 5, 6\}$ forms a group under the operation of multiplication modulo 7

(a) Show that G is cyclic.

(2)

The set $H = \{1, 5, 7, 11, 13, 17\}$ forms a group under the operation of multiplication modulo 18

(b) List all the subgroups of H .

(3)

(c) Describe an isomorphism between G and H .

(3)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 12 marks)



7. A transformation from the z -plane to the w -plane is given by

$$w = \frac{3iz - 2}{z + i} \quad z \neq -i$$

- (a) Show that the circle C with equation $|z + i| = 1$ in the z -plane is mapped to a circle D in the w -plane, giving a Cartesian equation for D .

(4)

- (b) Sketch C and D on Argand diagrams.

(2)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 6 marks)



8.

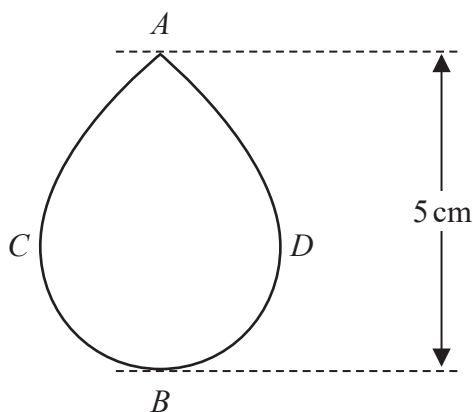


Figure 1

Figure 1 shows the vertical cross section of a child's spinning top. The point A is vertically above the point B and the height of the spinning top is 5 cm.

The line CD is perpendicular to AB such that CD is the maximum width of the spinning top.

The spinning top is modelled as the solid of revolution created when part of the curve with polar equation

$$r^2 = 25 \cos 2\theta$$

is rotated through 2π radians about the initial line.

(a) Show that, according to the model, the surface area of the spinning top is

$$k\pi(2 - \sqrt{2})\text{ cm}^2$$

where k is a constant to be determined.

(7)

(b) Show that, according to the model, the length CD is $\frac{5\sqrt{2}}{2}$ cm.

(6)



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 is 13 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

--	--	--	--	--

Candidate Number

--	--	--	--	--

Mock Paper Set 1

Time: 1 hour 30 minutes

Paper Reference **9FM0/4A**

Further Mathematics

Advanced

Paper 4A: Further Pure Mathematics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

S72166A

©2021 Pearson Education Ltd.

1/1/1/



S 7 2 1 6 6 A 0 1 2 8


Pearson

1.

$$\mathbf{A} = \begin{pmatrix} 1 & k \\ 3 & 4 \end{pmatrix} \quad \text{where } k \text{ is a constant}$$

(a) Use the Cayley-Hamilton theorem to show that

$$\mathbf{A}^3 = p\mathbf{A} + q\mathbf{I}$$

where $\mathbf{I}$ is the 2×2 identity matrix and p and q are constants in terms of k .

(3)

(b) Hence, determine the value of k such that $\mathbf{A}^3$ is diagonal.

(1)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2.

In this question you must show detailed reasoning.

Solve the congruence equations below.

In each case give your answers as the least positive residues for the given modulus.

(a) $6x \equiv 5 \pmod{11}$ (2)

(b) $6x \equiv 15 \pmod{33}$ (3)

(c) $4x \equiv 8^{343} \pmod{11}$ (4)



DO NOT WRITE IN THIS AREA

Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 9 marks)



3.

$$\mathbf{M} = \begin{pmatrix} 3 & 6 & 1 \\ p & 1 & 3 \\ -4 & q & 7 \end{pmatrix} \quad \text{where } p \text{ and } q \text{ are constants}$$

Given that $\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$ is an eigenvector of $\mathbf{M}$,

- (a) (i) show that $p = -6$
(ii) determine the corresponding eigenvalue.

(3)

Given also that another eigenvalue of $\mathbf{M}$ is 2

(b) find

- (i) the value of q ,
(ii) a corresponding eigenvector.

(4)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 7 marks)



4. (i) (a) Determine the number of different arrangements of the 13 letters

L O V E H O P E P E A C E

(2)

- (b) Determine how many arrangements there are in which the two P's do not appear next to each other.

(2)

- (ii) Determine how many numbers between 1 and 1000 contain the digit 7

- (a) at least once,

(2)

- (b) exactly once.

(2)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 8 marks)



5.

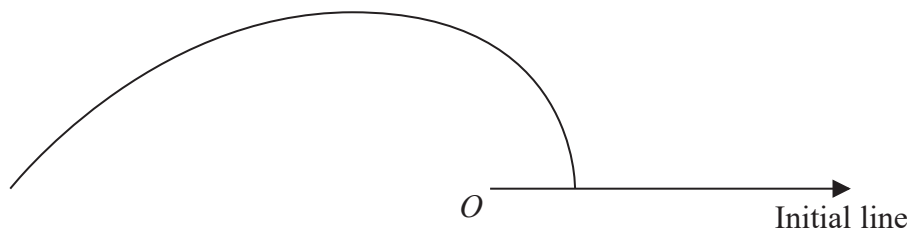
**Figure 1**

Figure 1 shows the curve defined by the polar equation

$$r = \sqrt{2} + \cosh \theta \quad 0 \leq \theta \leq \pi$$

(a) Show that

$$r^2 + \left(\frac{dr}{d\theta} \right)^2 = (1 + \sqrt{2} \cosh \theta)^2 \quad (4)$$

(b) Hence determine the length of this curve, giving your answer in the form $A + B \sinh C$, where A , B and C are real numbers to be found.

(3)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 7 marks)



6. An angling club models the number of fish in its lake, from when it opened, as follows:

- At the start of the first month there were 5000 fish in the lake
- The number of fish in the lake increases naturally by 6% each month
- At the **end** of each month 375 fish are removed from the lake by fishing

Let u_n be the number of fish in the lake at the start of the n th month.

(a) Explain why the model gives the recurrence system

$$u_1 = 5000 \quad u_{n+1} = 1.06u_n - 375 \quad (n = 1, 2, 3, \dots) \quad (2)$$

The substitution $v_k = u_k - \alpha$, where α is a constant, can be used to write the recurrence system in the form

$$v_1 = A \quad v_{n+1} = 1.06v_n \quad (n = 1, 2, 3, \dots)$$

where A is a constant.

(b) Determine

- (i) the value of α
 - (ii) the value of A
- (3)

(c) Hence determine a closed form for u_n

(3)

(d) Determine, according to the model, the number of months it takes for the lake to run out of fish.

(2)

If the model assumes instead that the 375 fish are removed from the lake at the **start** of each month,

- (e) (i) explain how the recurrence relation would change,
- (ii) state the effect this would have on the answer to part (d), giving a reason for your answer.
- (2)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 12 marks)



7.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

$$I_n = \int \frac{\cos nx}{\cos x} dx \quad n \geq 1$$

(a) Show that, for $n \geq 1$

$$I_{n+2} = \frac{2 \sin(n+1)x}{n+1} - I_n \quad (6)$$

(b) Hence determine the exact value of

$$\int_0^{\frac{\pi}{12}} \frac{\cos 5x}{\cos x} dx \quad (4)$$



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 10 marks)



8. The point P represents a complex number $z = x + iy$ in the complex plane such that

$$\arg\left(\frac{z-i}{z+2}\right) = \frac{\pi}{4}$$

The locus of P as z varies is an arc of a circle.

- (a) Determine a Cartesian equation for this circle in the form $(x - a)^2 + (y - b)^2 = r^2$ where a , b and r are constants. **(4)**
- (b) Sketch, on an Argand diagram, the locus of P as z varies.

- (b) Sketch, on an Argand diagram, the locus of P as z varies. (2)

The transformation T from the z -plane to the w -plane is given by

$$w = \frac{z + 3}{z - \mathbf{j}} \quad z \neq \mathbf{j}$$

The transformation T maps the locus of P onto a half-line in the w -plane.

- (c) Determine an equation for this half-line, giving your answer in Cartesian form and making clear which half of the line is the image of P
- (3)**

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 9 marks)



DO NOT WRITE IN THIS AREA

Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 9 is 9 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Candidate Number

--	--	--	--	--

--	--	--	--

Pearson Edexcel Level 3 GCE

Time 1 hour 30 minutes

Paper
reference

9FM0/4A

Further Mathematics

Advanced

PAPER 4A: Further Pure Mathematics 2

Mock Set 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

S73980A

©2022 Pearson Education Ltd.

1/1/1/



Pearson

1. (i) Using divisibility tests, determine if the number

1 188 810 197 253

(a) is divisible by 9

(2)

(b) is divisible by 11

(2)

- (ii) Given that 811 is a prime number, solve the congruence equation

$$x^{811} + 813x - 17 \equiv 0 \pmod{811}$$

(4)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 8 marks)



2. Determine a closed form for the recurrence system

$$u_1 = 3 \quad u_2 = -1$$

$$u_{n+2} = 3u_{n+1} + 4u_n - 12n \quad (n = 1, 2, 3, \dots)$$

(7)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 7 marks)



3. The Cayley tables below show two binary operations, $\oplus$ and $*$, on the set $G = \{a, b, c, d, e, f\}$

$\oplus$	<i>e</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>f</i>
<i>e</i>	<i>e</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>f</i>
<i>a</i>	<i>a</i>	<i>e</i>	<i>c</i>	<i>f</i>	<i>b</i>	<i>d</i>
<i>b</i>	<i>b</i>	<i>c</i>	<i>e</i>	<i>d</i>	<i>f</i>	<i>a</i>
<i>c</i>	<i>c</i>	<i>f</i>	<i>d</i>	<i>a</i>	<i>e</i>	<i>b</i>
<i>d</i>	<i>d</i>	<i>b</i>	<i>f</i>	<i>e</i>	<i>a</i>	<i>c</i>
<i>f</i>	<i>f</i>	<i>d</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>e</i>

$*$	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>
<i>a</i>	<i>e</i>	<i>f</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>
<i>b</i>	<i>d</i>	<i>c</i>	<i>b</i>	<i>a</i>	<i>f</i>	<i>e</i>
<i>c</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>
<i>d</i>	<i>f</i>	<i>e</i>	<i>d</i>	<i>c</i>	<i>b</i>	<i>a</i>
<i>e</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>a</i>	<i>b</i>
<i>f</i>	<i>b</i>	<i>a</i>	<i>f</i>	<i>e</i>	<i>d</i>	<i>c</i>

- (a) Use the elements a , b and c to show that the operation $\oplus$ is not associative.

(3)

Given that the operation $*$ is associative,

- (b) explain why $(G, *)$ forms a group,

(3)

- (c) determine a subgroup of $(G, *)$ of order 3

(1)

$(G, *)$ is isomorphic to one of the following two groups

- $(\mathbb{Z}_6, +_6)$, the set of integers modulo 6 with operation addition modulo 6
- $(S(\Delta), \circ)$, the set of symmetries of an equilateral triangle with operation of composition

- (d) Determine which one of these groups $(G, *)$ is isomorphic to, giving a reason why it is not isomorphic to the other group.

(2)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 9 marks)



4. A transformation T from the z -plane, where $z = x + iy$, to the w -plane, where $w = u + iv$, is given by

$$w = \frac{3}{z + j}$$

A line l in the z -plane has equation $z = x + ki$ where k is a constant, $k \neq -1$

- (a) Show that the line l maps to a circle in the w -plane, determining an equation for the circle in the form $(u - a)^2 + (v - b)^2 = r^2$ where a , b and r are constants in terms of k
- (5)

The region R in the z -plane is the region between the lines with equations

$$z = x + \frac{1}{2}i \quad \text{and} \quad z = x + 2i$$

The region S in the w -plane is the image of R under T .

- (b) On an Argand diagram sketch the region S . (5)

This image shows a full page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, typical of notebook paper. There are no margins, text, or other markings on the page.

DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 10 marks)



5.

$$\mathbf{M} = \begin{pmatrix} 6 & c & 3 \\ a & -5 & a \\ 7 & b & 10 \end{pmatrix} \quad \text{where } a, b \text{ and } c \text{ are constants}$$

- (a) Show that $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ is an eigenvector of $\mathbf{M}$, stating the corresponding eigenvalue.

Given that $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$ is another eigenvector for this eigenvalue (2)

- (b) determine the value of each of a , b and c . (3)

Given that the characteristic equation of $\mathbf{M}$ is

$$\lambda^3 - 11\lambda^2 + 39\lambda + k = 0$$

where k is a constant,

- (c) determine the other eigenvalue of $\mathbf{M}$ (2)

- (d) Determine a matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{D} = \mathbf{P}^{-1} \mathbf{M} \mathbf{P} \quad (4)$$

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 11 marks)



6.

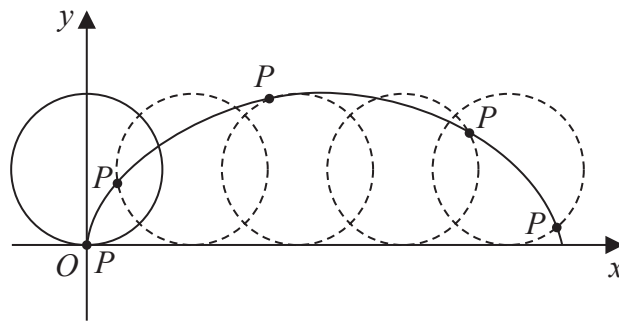


Figure 1

A cyclist cycled over a pin on the road and the pin stuck in the tyre of the front wheel of their bicycle.

The wheel remains in contact with the ground in subsequent motion.

Modelling this wheel as a circle moving in a vertical plane, the motion of the pin, P , for one revolution of the wheel is shown in Figure 1.

The path of P , with units in centimetres, is modelled by the equations

$$x = a(t - \sin t) \quad y = a(1 - \cos t) \quad t \geq 0$$

Given that the diameter of the wheel is 62 cm

- (a) determine the value of a (2)
- (b) Use algebraic integration to determine the total distance in metres travelled by P in one revolution of the wheel. (6)
- (c) Suggest a reason why the model may not remain suitable as the bicycle continues its motion. (1)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

7.

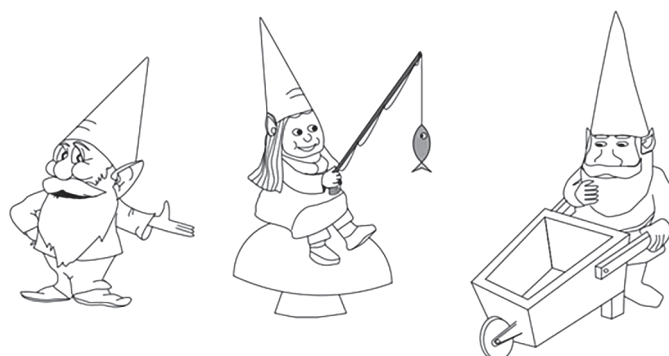


Figure 2

A garden store sells three designs of ornamental garden gnomes, shown in Figure 2.

The hat of each gnome is coloured either red, green or blue.

The clothes of each gnome are coloured either red, green or yellow.

No gnome has the same colour for its hat as it has for its clothes.

Two gnomes are considered the same if they have the same design, the same colour hat **and** the same colour clothes. They are considered different otherwise.

(a) Determine the total number of possible different gnomes that the garden store sells.

(2)

The store displays a row of 3 different gnomes each with a blue hat.

(b) Determine the number of possible arrangements of gnomes for this display.

(3)

A customer orders a set of 5 different gnomes.

(c) Determine the total number of possible sets of 5 gnomes.

(1)

The customer requires at least one of each of the three designs shown in Figure 2 in their set of gnomes.

(d) Determine the number of possible sets of 5 gnomes that satisfy the customer's requirement.

(3)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 9 marks)



8.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Determine $\frac{d}{dx}(\cos(\pi e^x))$

$$I_n = \int_0^{\ln 2} e^{nx} \sin(\pi e^x) \, dx \quad n \geq 1$$

(b) Show that, for $n \geq 3$

$$I_n = -\frac{2^{n-1} + 1}{\pi} - \frac{(n-1)(n-2)}{\pi^2} I_{n-2} \quad (6)$$

(c) Hence show that the exact value of I_5 is given by

$$I_5 = \frac{A}{\pi} + \frac{B}{\pi^3} + \frac{C}{\pi^5}$$

where A , B and C are integers to be determined. (4)



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



Write your name here

Surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

--	--	--	--	--

Candidate Number

--	--	--	--	--

Further Mathematics

Advanced

Further Mathematics Option 2

Paper 4: Further Pure Mathematics 2

Sample Assessment Material for first teaching September 2017

Time: 1 hour 30 minutes

Paper Reference

9FM0/4A

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B)
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

S54444A

©2017 Pearson Education Ltd.

1/1/1/1/



S 5 4 4 4 4 4 A 0 1 2 9



Pearson

Answer ALL questions. Write your answers in the spaces provided.

1. (i) Use the Euclidean algorithm to find the highest common factor of 602 and 161.

Show each step of the algorithm.

(3)

- (ii) The digits which can be used in a security code are the numbers 1, 2, 3, 4, 5, 6, 7, 8 and 9.

Originally the code used consisted of two distinct odd digits, followed by three distinct even digits.

To enable more codes to be generated, a new system is devised. This uses two distinct even digits, followed by any three other distinct digits. No digits are repeated.

Find the increase in the number of possible codes which results from using the new system.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 3 Advanced GCE in Mathematics
Sample Assessment Materials – Issue 1 – June 2017 © Pearson Education Limited 2017

2. A transformation from the z -plane to the w -plane is given by

$$w = z^2$$

- (a) Show that the line with equation $\text{Im}(z) = 1$ in the z -plane is mapped to a parabola in the w -plane, giving an equation for this parabola.

(4)

- (b) Sketch the parabola on an Argand diagram.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 3 Advanced GCE in Mathematics
Sample Assessment Materials – Issue 1 – June 2017 © Pearson Education Limited 2017

3. The matrix $\mathbf{M}$ is given by

$$\mathbf{M} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ -1 & 0 & 4 \end{pmatrix}$$

(a) Show that 4 is an eigenvalue of $\mathbf{M}$, and find the other two eigenvalues.

(4)

(b) For each of the eigenvalues find a corresponding eigenvector.

(4)

(c) Find a matrix $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{M}\mathbf{P}$ is a diagonal matrix.

(2)

DO NOT WRITE IN THIS AREA

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 3 continued

(Total for Question 3 is 10 marks)

4. (i) A group G contains distinct elements a , b and e where e is the identity element and the group operation is multiplication.

Given $a^2b = ba$, prove $ab \neq ba$

(4)

- (ii) The set $H = \{1, 2, 4, 7, 8, 11, 13, 14\}$ forms a group under the operation of multiplication modulo 15

(a) Find the order of each element of H .

(3)

(b) Find three subgroups of H each of order 4, and describe each of these subgroups.

(4)

The elements of another group J are the matrices
$$\begin{pmatrix} \cos\left(\frac{k\pi}{4}\right) & \sin\left(\frac{k\pi}{4}\right) \\ -\sin\left(\frac{k\pi}{4}\right) & \cos\left(\frac{k\pi}{4}\right) \end{pmatrix}$$

where $k = 1, 2, 3, 4, 5, 6, 7, 8$ and the group operation is matrix multiplication.

(c) Determine whether H and J are isomorphic, giving a reason for your answer.

(2)

DO NOT WRITE IN THIS AREA

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 3 Advanced GCE in Mathematics
Sample Assessment Materials – Issue 1 – June 2017 © Pearson Education Limited 2017

5.

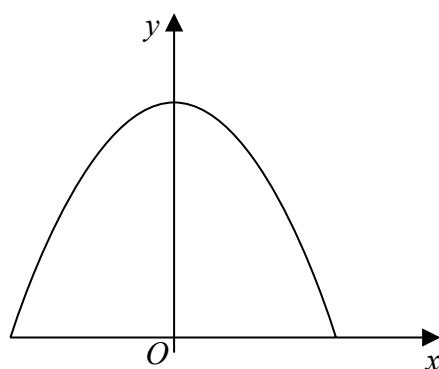


Figure 1

An engineering student makes a miniature arch as part of the design for a piece of coursework.

The cross-section of this arch is modelled by the curve with equation

$$y = A - \frac{1}{2} \cosh 2x, \quad -\ln a \leq x \leq \ln a$$

where $a > 1$ and A is a positive constant. The curve begins and ends on the x -axis, as shown in Figure 1.

- (a) Show that the length of this curve is $k \left(a^2 - \frac{1}{a^2} \right)$, stating the value of the constant k . (5)

The length of the curved cross-section of the miniature arch is required to be 2 m long.

- (b) Find the height of the arch, according to this model, giving your answer to 2 significant figures. (4)

- (c) Find also the width of the base of the arch giving your answer to 2 significant figures. (1)

- (d) Give the equation of another curve that could be used as a suitable model for the cross-section of an arch, with approximately the same height and width as you found using the first model.
(You do not need to consider the arc length of your curve) (2)

DO NOT WRITE IN THIS AREA

Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 3 Advanced GCE in Mathematics
Sample Assessment Materials – Issue 1 – June 2017 © Pearson Education Limited 2017

6. A curve has equation

$$|z + 6| = 2 |z - 6| \quad z \in \mathbb{C}$$

(a) Show that the curve is a circle with equation $x^2 + y^2 - 20x + 36 = 0$ (2)

(b) Sketch the curve on an Argand diagram. (2)

The line l has equation $az^* + a^*z = 0$, where $a \in \mathbb{C}$ and $z \in \mathbb{C}$

Given that the line l is a tangent to the curve and that $\arg a = \theta$

(c) find the possible values of $\tan \theta$ (5)

DO NOT WRITE IN THIS AREA

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 3 Advanced GCE in Mathematics
Sample Assessment Materials – Issue 1 – June 2017 © Pearson Education Limited 2017

7.

$$I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx, \quad n \geq 0$$

(a) Prove that, for $n \geq 2$,

$$nI_n = (n-1)I_{n-2} \quad (4)$$

(b)

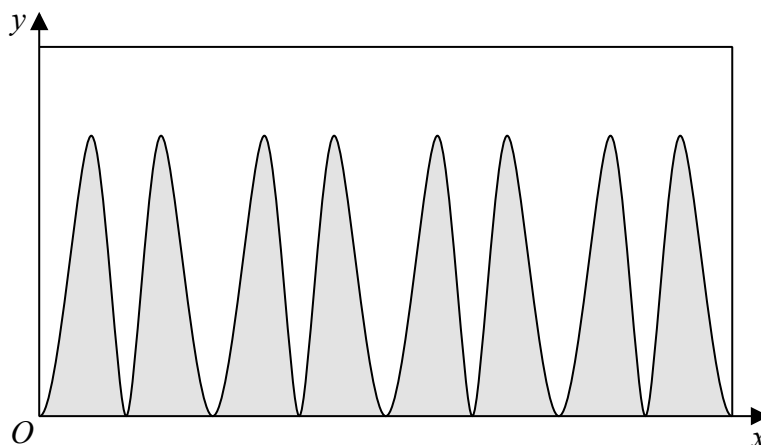


Figure 2

A designer is asked to produce a poster to completely cover the curved surface area of a solid cylinder which has diameter 1 m and height 0.7 m.

He uses a large sheet of paper with height 0.7 m and width of π m.

Figure 2 shows the first stage of the design, where the poster is divided into two sections by a curve.

The curve is given by the equation

$$y = \sin^2(4x) - \sin^{10}(4x)$$

relative to axes taken along the bottom and left hand edge of the paper.

The region of the poster below the curve is shaded and the region above the curve remains unshaded, as shown in Figure 2.

Find the exact area of the poster which is shaded.

(5)

DO NOT WRITE IN THIS AREA

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

8. A staircase has n steps. A tourist moves from the bottom (step zero) to the top (step n). At each move up the staircase she can go up either one step or two steps, and her overall climb up the staircase is a combination of such moves.

If u_n is the number of ways that the tourist can climb up a staircase with n steps,

- (a) explain why u_n satisfies the recurrence relation

$$u_n = u_{n-1} + u_{n-2}, \text{ with } u_1 = 1 \text{ and } u_2 = 2 \quad (3)$$

- (b) Find the number of ways in which she can climb up a staircase when there are eight steps.

(1)

A staircase at a certain tourist attraction has 400 steps.

- (c) Show that the number of ways in which she could climb up to the top of this staircase is given by

$$\frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^{401} - \left(\frac{1 - \sqrt{5}}{2} \right)^{401} \right] \quad (5)$$

DO NOT WRITE IN THIS AREA

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

TOTAL FOR PAPER IS 75 MARKS

Please check the examination details below before entering your candidate information

Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

--	--	--	--	--

Candidate Number

--	--	--	--	--

Specimen Paper

(Time: 1 hour 30 minutes)

Paper Reference **9FM0/4A**

Further Mathematics

Advanced

Paper 4A: Further Pure Mathematics 2

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B)
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

S61300A

©2018 Pearson Education Ltd.

1/1/1/1/




Pearson

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- DO NOT WRITE IN THIS AREA**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(3)

(2)

- 2



Question 1 continued

Handwriting practice area with 30 horizontal lines.

(Total for Question 1 is 5 marks)



$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 5 & 4 \\ 4 & 4 & 3 \end{pmatrix}$$

- (3)

- (5)

(c) write down a matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that

(4)

DONOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



S 6 1 3 0 0 A 0 5 2 4

Question 2 continued

Lined area for writing the answer to Question 2.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 12 marks)



3. (i) Lottery X requires each player to buy a ticket and choose 5 different numbers from the numbers 1 to 45 inclusive.

Lottery Y requires each player to buy a ticket and choose 6 different numbers from the numbers 1 to 35 inclusive.

A player wins if their chosen numbers match completely those drawn at random by the lotteries.

A person wishes to play one of these two lotteries.

The price of a ticket to play each lottery is the same.

The prize money for winning each lottery is the same.

Decide which lottery you would recommend that they play, giving a reason for your answer.

(2)

(ii) Use Fermat's little theorem to show that when 128^{129} is divided by 17 the remainder is 9

(4)

(iii) There are $3x$ chairs in a room. When these chairs are set out in rows of 7 there are two chairs left over.

(a) Form and solve a congruence equation for x

(3)

Given that there are at least 100 chairs and that one third of the chairs can be arranged exactly into 5 equal rows,

(b) find the least possible number of chairs in the room.

(3)



Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 12 marks)



4. (i) Two distinct elements of a group G are a and b .
The element a has order 5 and $a^3b = ba^3$
Prove that $ab = ba$

(4)

Given that p, q, r and s are distinct elements

- (ii) (a) check each of the group axioms for the set $A = \{p, q, r, s\}$ under the operation $\oplus$ defined in the table below.
- (b) Hence determine whether the set A forms a group under the operation $\oplus$.

$\oplus$	p	q	r	s
p	p	q	r	s
q	q	p	q	r
r	r	q	p	q
s	s	r	q	p

(3)



Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 7 marks)



5.

$$I_n = \int_0^{\frac{\pi}{2}} x^n \cos x \, dx, \quad n \geq 0$$

(a) Prove that, for $n \geq 2$

$$I_n = \left(\frac{\pi}{2}\right)^n - n(n-1)I_{n-2} \quad (5)$$

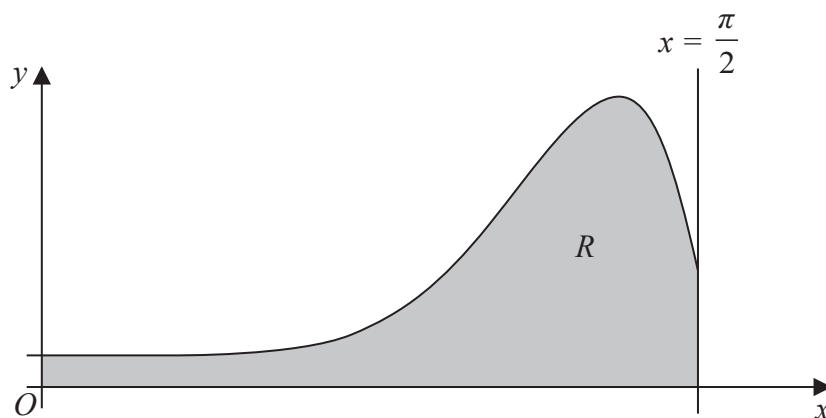


Figure 1

Figure 1 shows the vertical cross-section of a proposed flood defence system. The cross-section of the flood defence system is modelled by the curve with equation

$$y = 1.2x^6 \cos x + 0.2 \quad 0 \leq x \leq \frac{\pi}{2}$$

where x and y are measured in metres.

The area R , shown shaded in Figure 1, is bounded by the curve, the y -axis, the x -axis and the line with equation $x = \frac{\pi}{2}$

The flood defence system will come in hollow sections that will be filled with water once they are in place. Each section will have a length of 10 metres.

(b) Use the model and the answer to part (a), to estimate the volume of water needed to fill each section. (6)

Each section can be filled with water at a maximum rate of 175 litres per minute and is required to be filled with water within 1 hour of being put in place.

(c) Use the model to decide whether this requirement can be met, showing all your reasoning. (2)



Question 5 continued

Lined area for writing the answer to Question 5.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 5 continued

Lined area for writing the answer to Question 5.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 13 marks)



DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- (7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- (6)

Question 6 continued

Lined area for writing the answer to Question 6.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 6 continued

Lined area for writing the answer to Question 6.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 13 marks)



7. (i) Solve the recurrence relation

$$U_{n+2} = 6U_{n+1} - 9U_n + 4 \quad n \geq 1$$

$$U_1 = 4 \qquad U_2 = 7$$

(7)

(ii) A sequence $x_1, x_2, x_3, \dots$ is defined by

$$(n+1)x_{n+2} + x_{n+1} - (n+2)x_n = 0 \quad n \geq 1$$

$$x_1 = 2 \qquad x_2 = 5$$

Prove by induction that, for $n \in \mathbb{Z}^+$

$$x_n = \frac{1}{8}[25 + 3(1 + 2n)(-1)^n]$$

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 7 continued

Handwriting practice area with 20 horizontal lines.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 7 continued

Lined area for writing the answer to Question 7.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 7 continued

Lined area for writing the answer to Question 7.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 13 marks)

TOTAL FOR PAPER IS 75 MARKS

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

